Electromagnetic Waves

Problem 9.1

By explicit differentiation, check that the functions $f_1, f_2,$ and f_3 satisfy the wave equation. Show that f_4 and f_5 do not.

$$
\begin{aligned}
f_1(z,t) &= Ae^{-b(z-vt)^2} \\
f_2(z,t) &= A\sin[b(z-vt)] \\
f_3(z,t) &= \frac{A}{b(z-vt)^2+1} \\
f_4(z,t) &= Ae^{-b(bz^2+vt)} \\
f_5(z,t) &= A\sin(bz)\cos(bvt)^3
\end{aligned}
$$

Solution:

The wave equation is

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}.$$

f_1:

$$
\begin{aligned}
\frac{\partial f_1}{\partial z} &= -2bA(z-vt)e^{-b(z-vt)^2}, & \frac{\partial^2 f_1}{\partial z^2} &= 4b^2A(z-vt)^2e^{-b(z-vt)^2} \\
\frac{\partial f_1}{\partial t} &= 2vbA(z-vt)e^{-b(z-vt)^2}, & \frac{\partial^2 f_1}{\partial t^2} &= 4v^2b^2A(z-vt)^2e^{-b(z-vt)^2}
\end{aligned}
$$

$$\boxed{\begin{gathered}\frac{\partial^2 f_1}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f_1}{\partial t^2} \\ \downarrow \\ 4b^2A(z-vt)^2e^{-b(z-vt)^2} = \frac{1}{v^2}4v^2b^2A(z-vt)^2e^{-b(z-vt)^2}\checkmark\end{gathered}}$$

f_2:

$$\frac{\partial f_2}{\partial z} = bA\cos[b(z-vt)], \qquad \frac{\partial^2 f_2}{\partial z^2} = -b^2 A\sin[b(z-vt)]$$
$$\frac{\partial f_2}{\partial t} = -vbA\cos[b(z-vt)], \qquad \frac{\partial^2 f_2}{\partial t^2} = -v^2 b^2 A\sin[b(z-vt)]$$

$$\boxed{\begin{gathered}\frac{\partial^2 f_2}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f_2}{\partial t^2} \\ \downarrow \\ -b^2 A\sin[b(z-vt)] = -\frac{1}{v^2}v^2 b^2 A\sin[b(z-vt)]\checkmark\end{gathered}}$$

f_3 using quotient rule:

$$\frac{\partial f_3}{\partial z} = \frac{-2bA(z-vt)}{\left(b(z-vt)^2+1\right)^2}$$
$$\frac{\partial^2 f_3}{\partial z^2} = \frac{-2bA}{\left(b(z-vt)^2+1\right)^2} - \frac{-2bA(z-vt)2\left(b(z-vt)^2+1\right)(2b(z-vt))}{\left(b(z-vt)^2+1\right)^4} =$$
$$= \frac{-2bA}{\left(b(z-vt)^2+1\right)^2} + \frac{8b^2A(z-vt)^2}{\left(b(z-vt)^2+1\right)^3}$$
$$\frac{\partial f_3}{\partial t} = \frac{2bvA(z-vt)}{\left(b(z-vt)^2+1\right)^2}$$
$$\frac{\partial^2 f_3}{\partial t^2} = \frac{-2bv^2A}{\left(b(z-vt)^2+1\right)^2} - \frac{2bvA(z-vt)2\left(b(z-vt)^2+1\right)(-2bv(z-vt))}{\left(b(z-vt)^2+1\right)^4} =$$
$$= \frac{-2bv^2A}{\left(b(z-vt)^2+1\right)^2} + \frac{8b^2v^2A(z-vt)^2}{\left(b(z-vt)^2+1\right)^3}$$

$$\boxed{\frac{\partial^2 f_3}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f_3}{\partial t^2}\checkmark}$$

f_4:

$$\frac{\partial f_4}{\partial z} = -2b^2 zAe^{-b(bz^2+vt)}, \qquad \frac{\partial^2 f_4}{\partial z^2} = (-2b^2+4b^4z^2)\left(Ae^{-b(bz^2+vt)}\right)$$
$$\frac{\partial f_4}{\partial t} = -bvAe^{-b(bz^2+vt)}, \qquad \frac{\partial^2 f_4}{\partial t^2} = b^2v^2Ae^{-b(bz^2+vt)}$$

$$\boxed{\frac{\partial^2 f_4}{\partial z^2} \neq \frac{1}{v^2}\frac{\partial^2 f_4}{\partial t^2}\times}$$

f_5:

$$\frac{\partial f_5}{\partial z} = bA\cos(bz)\cos(bvt)^3, \quad \frac{\partial^2 f_5}{\partial z^2} = -b^2 A\sin(bz)\cos(bvt)^3$$
$$\frac{\partial f_5}{\partial t} = -3(b^3v^3t^2)\sin(bz)\sin(bvt)^3$$
$$\frac{\partial^2 f_5}{\partial t^2} = -9(b^6v^6t^4)\sin(bz)\cos(bvt)^3 - 6(b^3v^3t)\sin(bz)\sin(bvt)^3$$

$$\boxed{\frac{\partial^2 f_5}{\partial z^2} \neq \frac{1}{v^2}\frac{\partial^2 f_5}{\partial t^2} \times}$$

□

Problem 9.2

Show that the **standing wave** $f(z,t) = A\sin(kz)\cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right.

Solution:

$$\frac{\partial^2 f}{\partial z^2} = -k^2 A\sin(kz)\cos(kvt)$$
$$\frac{\partial^2 f}{\partial t^2} = -v^2k^2 A\sin(kz)\cos(kvt)$$

The wave equation

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}$$

is satisfied. Using the trigonometric identity

$$\sin(a)\cos(b) = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$$

we obtain

$$\frac{A}{2}[\sin(k(z+vt)) + \sin(k(z-vt))]$$

which is the sum of a wave traveling to the right and left. □

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Problem 9.4

Problem 9.5

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